

# Algorithm Lab Project

**by:**

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## Description of code:

Find smallest number of two arrays

You are given two unsorted arrays of integers a and b of the same sizes, sort them using two different sorting (Merge Sort & Quick Sort) algorithms then put the smallest number in the same index in third array c (Note: The numbers should be different in array a and b). Then Search about number in array c and output the number is found in index i in array a or b. Print not Found if the number is not exist in both arrays. Then calculate Big (O) of your program.

Input

- The second line contains n integers — the elements of array a.
- The third line contains m integers — the elements of array b.
- Search in the output array.

Output

- Array c which contains the smallest number between array a and b in the same index.
- Result of searching (Found at any index in which array /or Not Found).

## Run of code without GUI:

```
Output - Run (FinalAlgorithmProject)
-----<C:\com.mycompany\FinalAlgorithmProject >-----
Building FinalAlgorithmProject 1.0-SNAPSHOT
from pom.xml
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ FinalAlgorithmProject ---
skip non existing resourceDirectory E:\aBooks of College\second year second term\an
--- compiler:3.11.0:compile (default-compile) @ FinalAlgorithmProject ---
Changes detected - recompiling the module! :source
Compiling 1 source file with javac [debug target 21] to target\classes
--- exec:3.1.0:exec (default-cli) @ FinalAlgorithmProject ---
Enter elements of array a separated by spaces:
3 2 5 1 7
Enter elements of array b separated by spaces (must same size array a):
8 4 6 10 9
Array c: [1, 2, 3, 5, 7]
Enter Search Key:
3
3 is found at index 2 in array a

The time complexity of Merge Sort is O(n log n)
The time complexity of Quick Sort is O(n^2)
The time complexity of the binarySearch function is O(log n)
The Big O complexity of the entire program is O(n^2)
-----
BUILD SUCCESS
-----
Total time: 41.351 s
Finished at: 2024-05-11T16:59:32+03:00
-----
```

## Run of code with GUI:

Final Algorithm Project

Array a:

8 4 2 6 9

Array b:

10 5 3 1 7

Search Key:

9

Array c: [ 1, 3, 5, 7, 9 ]

9 is found at index 4 in array a

The time complexity of Merge Sort is O(n log n)

The time complexity of Quick Sort is O(n^2)

The time complexity of the binarySearch function is O(log n)

The Big O complexity of the entire program is O(n^2)

Search